GI Acidity, Ulcers, and GERD Agents

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PHSC 2623: Pharmacologic Basis of Therapeutics II

Objectives

- Recall common acid-peptic disorders and their pathophysiology
- Identify key cells and receptors responsible for the production of gastric acid
- Describe the mechanism of action, adverse effects, and clinical utility of antacids
- Describe the mechanism of action, adverse effects, and clinical utility of Histamine$_2$–Receptor Antagonists
- Identify relevant drug interactions caused by Histamine$_2$–Receptor Antagonists

Acid-Peptic Disorders

- Include
  - Gastroesophageal Reflux Disease (GERD)
  - Peptic Ulcer Disease (PUD)
    - Gastric and Duodenal
  - Stress-related mucosal injury
- Caused when the caustic effects of acid, pepsin, and bile overwhelm the defensive factors of the GI mucosa (mucus, bicarbonate, prostaglandins, etc.)

Acid-Peptic Disorders

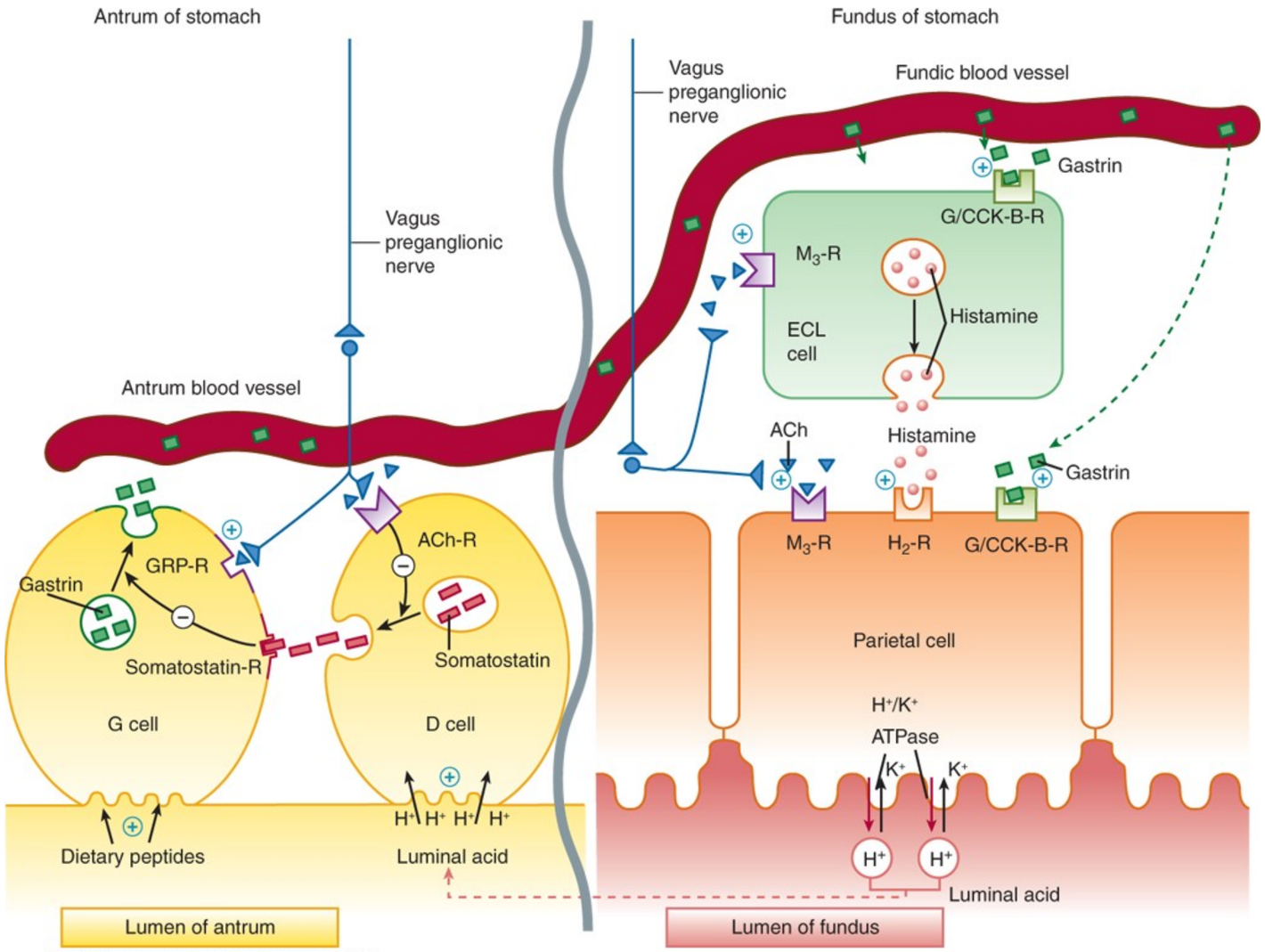
- >90% of peptic ulcers are caused by infection with *Helicobacter pylori* or by use of nonsteroidal anti-inflammatory drugs (NSAIDs)
- Pharmacologic agents that treat acid-peptic disorders can be divided into two classes:
  - Agents that reduce intragastric acidity
  - Agents that promote mucosal defense

Physiology of Acid Secretion

- Parietal cells of the stomach contain the following receptors
  - **CCK-B** (which binds **gastrin**)
    - Gastrin is released from **antral G cells** into the blood
  - **H_2** (which binds **histamine**)
  - **M_3** (which binds **acetylcholine**)
    - Acetylcholine is released from the **vagal postganglionic nerves**

Physiology of Acid Secretion

- Gastrin and acetylcholine bind to their respective parietal cell receptors
  - ⬆ cytosolic calcium ➡ stimulates protein kinases ➡ stimulates acid
  - secretion from a H^+/K^+-ATPase (proton) pump on the canalicular surface
- Adjacent to parietal cells are **Enterochromaffin-like (ECL) cells**
  - Also have receptors for gastrin and acetylcholine
    - When stimulated, facilitates the release of **histamine**
      - Binds to H_2 receptors on the parietal cell ➡ activates adenylyl cyclase ⬆ intracellular cyclic adenosine monophosphate (cAMP) ➡ activates protein kinases ➡ stimulates acid secretion by the H^+/K^+-ATPase



Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition
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Antacids

- Oldest treatment for patients with dyspepsia and acid-peptic disorders
- A mainstay of therapy until the advent of H_2-receptor antagonists and proton-pump inhibitors (PPIs)
- Continue to be used commonly by patients as OTC remedies for treatment of intermittent heartburn and dyspepsia
- Weak bases that react with **gastric HCl** acid to form **a salt and water**
- **Principal MOA**: reduction of intragastric acidity

Antacids

- Approximately 45 mEq/h of HCl acid are secreted post-prandially
- A single antacid dose of 156 mEq given an hour post-prandially can neutralize gastric acid for up to 2 hours
  - Acid neutralization capacity among various proprietary formulations of antacids is highly variable depending on their
    - Rate of dissolution (tablet vs liquid)
    - Water solubility
    - Rate of reaction with acid
    - Rate of gastric emptying

Antacids

- **Sodium bicarbonate** (**baking soda**, **Alka Seltzer**)
  - Reacts rapidly with HCl to produce CO_2 and NaCl
    - CO_2 can result in gastric distention, belching, and potentially metabolic alkalosis when given in high doses or to patients with renal insufficiency
    - NaCl may exacerbate fluid retention in patients with HF, HTN, or renal insufficiency

Antacids

- **Calcium carbonate** (**Tums**, **Os-Cal**)
  - Less soluble and reacts more slowly than sodium bicarbonate
  - Forms CO_2 and calcium chloride ($CaCl_2$)
  - May also cause belching or metabolic alkalosis
    - Hypercalcemia and renal inefficiency in excessive doses
  - Has several other indications (i.e. calcium replacement)

Antacids

- **Magnesium hydroxide** and **Aluminum hydroxide** formulations
  - React slowly with HCl to form **magnesium chloride** or **aluminum chloride** and water
  - No gas is generated: belching does not occur
  - **Unabsorbed magnesium salts** can cause **osmotic diarrhea**
  - **Unabsorbed aluminum salts** can cause **constipation**
    - Thus, these agents are commonly formulated together (Maalox, Mylanta)

Antacids

- **Magnesium hydroxide** and **Aluminum hydroxide** formulations
  - Both are absorbed or excreted by the kidneys
    - Should not be used long term in patients with renal insufficiency
- **All antacids may affect the absorption of other medications** by
  - Binding the drug (reducing its absorption)
  - Increasing intragastric pH so the drug's dissolution or solubility is altered
- No antacids should be administered within 2 hours of doses of **tetracyclines**, **fluroquinolones**, **itraconazole**, and **iron**

H_2-Receptor Antagonists

- Were the most commonly prescribed drugs in the world from the 1970s to 1990s
- Three are currently in clinical use: **cimetidine**, **famotidine**, and **nizatidine**
  - All are rapidly absorbed from the intestine
  - Cimetidine and famotidine undergo first-pass metabolism
    - Bioavailability of ~50%
  - Nizatidine has little first pass metabolism

H_2-Receptor Antagonists

- Serum half-lives range from **1.1 to 4 hours**
  - Duration of action depends on the administered dose
- Cleared by a combination of hepatic metabolism, glomerular filtration, and renal tubular secretion.
- Dose reduction is required in patients with moderate to severe renal (and possibly severe hepatic) insufficiency
- Decline of up to 50% in drug clearance as well as significant reduction in volume of distribution in the elderly

Bertram G. Katzung, Todd W. Vanderah+

TABLE 62–1Clinical comparisons of H_2-receptor blockers.

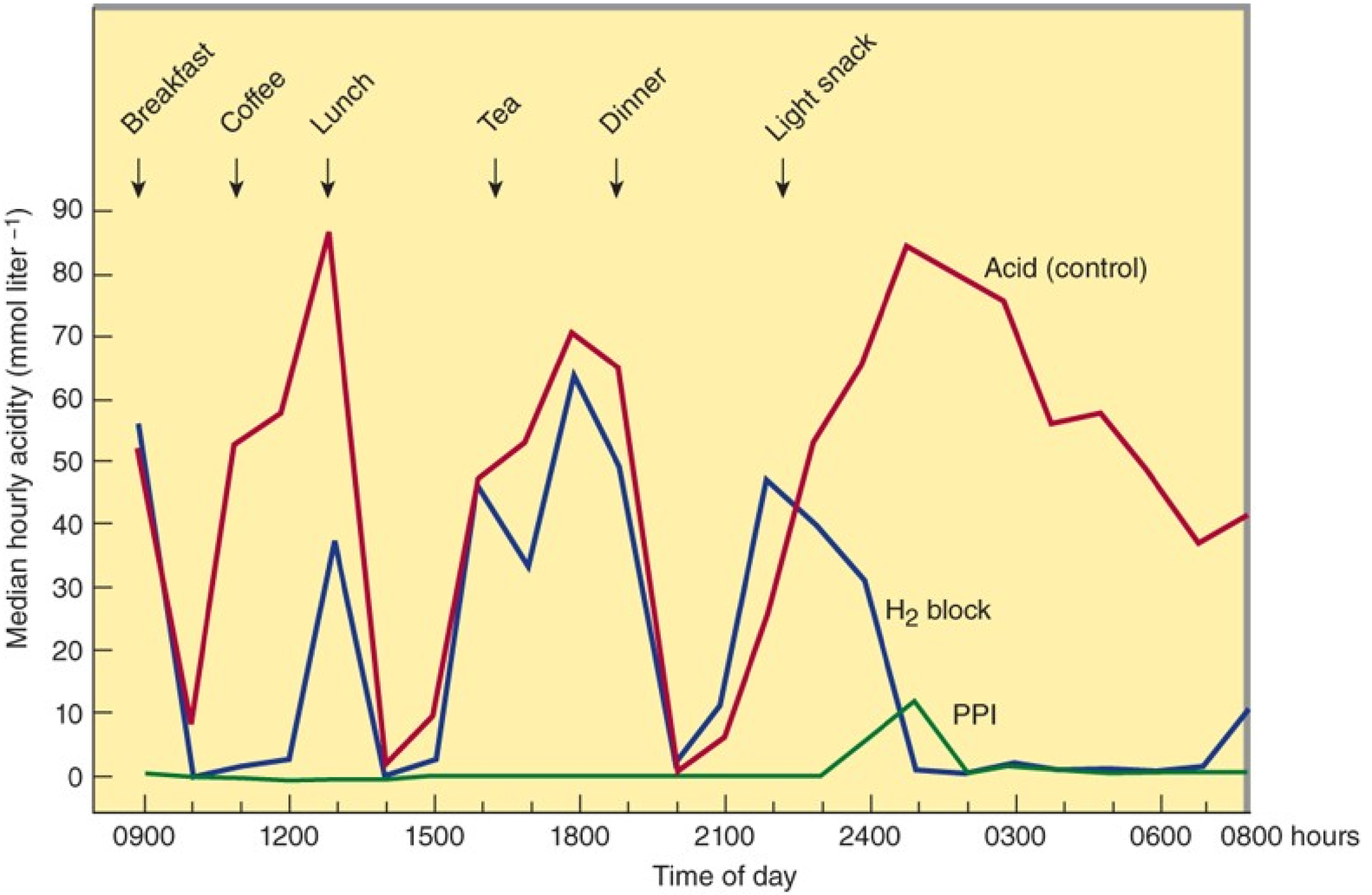
Drug	Relative Potency	Dose to Achieve >50% Acid Inhibition for 10 Hours	Usual Dose for Acute Duodenal or Gastric Ulcer	Usual Dose for Gastroesophageal Reflux Disease	Usual Dose for Prevention of Stress-Related Bleeding
Cimetidine	1	400–800 mg	800 mg HS or 400 mg bid	800 mg bid	50 mg/h continuous infusion
Ranitidine	4–10	150 mg	300 mg HS or 150 mg bid	150 mg bid	6.25 mg/h continuous infusion or 50 mg IV every 6–8 h
Nizatidine	4–10	150 mg	300 mg HS or 150 mg bid	150 mg bid	Not available
Famotidine	20–50	20 mg	40 mg HS or 20 mg bid	20 mg bid	20 mg IV every 12 h

bid, twice daily; HS, bedtime.

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H_2-Receptor Antagonists

- Exhibit competitive inhibition at the parietal cell H_2 receptor
- Suppress basal and meal stimulated acid secretion in a linear, dose-dependent manner
- Highly selective for H_2 (do not affect H_1 and H_3)
- Reduce overall gastric secretion and concentration of pepsin



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H_2-Receptor Antagonists

- Reduce acid secretion stimulated by histamine, gastrin, and cholinomimetic agents through 2 mechanisms
  - **Block histamine** released from ECL cells by gastrin or vagal stimulation from binding to the parietal H_2 receptors
  - H_2 receptor blockade **diminishes the direct parietal cell stimulation effect** of gastrin or acetylcholine on acid secretion
- Even with varying potencies, all H_2-Receptor Antagonists **inhibit 60-70% of total 24 hour acid secretion** (nocturnal > post-prandial)
  - Recommended prescription doses maintain >50% acid inhibition for 10 hours
    - OTC doses have a 6-10 hour acid inhibition

H_2-Receptor Antagonists

- **Clinical Uses: Gastroesophageal Reflux Disease (GERD)**
  - Patients w/infrequent heartburn or dyspepsia (<3 times per week) may take either antacids or intermittent H_2 antagonists
    - Duration of action: H_2 antagonists (6-10 hours) > Antacids (1-2 hours)
  - H_2 antagonists may be taken prophylactically before meals to reduce heartburn
  - Frequent heartburn is treated with BID H_2 antagonists or PPIs
  - In patients with erosive esophagitis (>50% of patients w/GERD), H_2 antagonists provide healing in <50% of patients

H_2-Receptor Antagonists

- **Clinical Uses: Peptic Ulcer Disease (PUD)**
  - Have largely been replaced by PPIs for treating acute PUD
  - Nocturnal acid suppression by H_2 antagonists provides effective ulcer healing in most patients with uncomplicated gastric and duodenal ulcers
  - When administered at bedtimes, provide **80-90% ulcer healing** rates after 6-8 weeks of therapy
  - Patients w/ ulcers caused by NSAIDs or aspirin should D/C the offending agent
    - If an NSAID must be continued for clinical reasons, a PPI should be used
  - No role in acute peptic ulcers caused by *H pylori* (treated w/abx +PPI)

H_2-Receptor Antagonists

- **Clinical Uses: Nonulcer dyspepsia**
  - Commonly used OTC and prescription agents for intermittent dyspepsia not caused by peptic ulcer
  - No benefit has been demonstrated when compared with placebo

H_2-Receptor Antagonists

- **Clinical Uses: Prevention of Bleeding from stress-related gastritis**
  - Agents that increase intragastric pH (H_2 antagonists and PPIs) reduce the risk of clinically significant bleeding and should be administered to patients who are at **high risk** of a GI bleed
    - Typically reserved for **critically ill** patients (reduced mobility)
      - 1-5% experience bleeding from upper GI erosions or ulcers
    - The optimal agent is uncertain (H_2 antagonists vs PPIs)
      - Both drug classes can be administered intravenously

H_2-Receptor Antagonists

- **Adverse effects**
  - Extremely well tolerated (<3% of patients experience adverse effects)
    - Diarrhea, headache, fatigue, myalgias, and constipation have been reported
  - Critically ill patients
    - Correlation between IV H_2 antagonists & the risk of **nosocomial pneumonia**
    - Mental status changes may occur in elderly patients or those with reduced hepatic or renal function (more common with cimetidine)
      - Rare in ambulatory patients

H_2-Receptor Antagonists

- **Adverse effects**
  - Cimetidine inhibits binding of dihydrotestosterone to androgen receptor, inhibits metabolism of estradiol, and increases serum prolactin levels
    - Long-term or high dose use may cause **gynecomastia or impotence in men** or **galactorrhea in women**
  - **Pregnancy/breastfeeding:** No harmful fetal effects but they do cross the placenta and are secreted in breast milk
    - Use should be minimized

H_2-Receptor Antagonists

- **Adverse effects**
  - May rarely cause blood dyscrasias
  - Blockade of cardiac H_2 may cause bradycardia (rarely clinically significant) or hypotension
    - Especially in rapid IV infusion
      - Should be given over 30 minutes
  - Rarely cause reversible abnormalities in liver chemistry

H_2-Receptor Antagonists

- **Drug Interactions**
  - Cimetidine interferes with several important hepatic cytochrome P450 pathways
    - **1A2, 2C9, 2D6, and 3A4**
      - Half-lives of drugs metabolized by these pathways may be prolonged when co-administered.
  - Nizatidine and Famotidine have negligible CYP interactions

H_2-Receptor Antagonists

- **Drug Interactions**
  - All compete with creatinine and certain other drugs (i.e. procainamide) for renal tubular secretion
  - Cimetidine and nizatidine inhibit first-pass metabolism of ethanol, especially in women
    - ⬆ bioavailability of ethanol could ⬆ blood ethanol levels
      - Clinical relevance of this is debated

Reference

Fantry GT. Drugs Used in the Treatment of Gastrointestinal Diseases. In: Vanderah TW. eds. *Katzung's Basic & Clinical Pharmacology, 16th Edition*. McGraw Hill; 2024. Accessed October 30, 2023. https://accesspharmacy.mhmedical.com/content.aspx?bookid=3382&sectionid=281757901

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